Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z4,ZL

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Contents

1 Examined $H(z)$ for CG TIA Automated NO L simple Z4 ZL: $\frac{Z_4 Z_L g_m}{Z_4 g_m + 2 Z_L g_m}$

$$H(z) = \frac{Z_4 Z_L g_m}{Z_4 g_m + 2 Z_L g_m}$$

2 AP

3 BP

4 BS

5 GE

6 HP

7 LP

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(\infty,\ \infty,\ \infty,\ \frac{R_4}{C_4 R_4 s + 1},\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_L R_4 R_L s + R_4}{2 C_4 C_L R_4 R_L s^2 + s\left(2 C_4 R_4 + C_L R_4 + 2 C_L R_L\right) + 2}$$

Parameters:

Q: $\frac{2\sqrt{C_4}\sqrt{C_L}\sqrt{R_4}\sqrt{R_L}}{2C_4R_4+C_LR_4+2C_LR_L}$
wo: $\frac{1}{\sqrt{C_4}\sqrt{C_L}\sqrt{R_4}\sqrt{R_L}}$
bandwidth: $\frac{2C_4R_4+C_LR_4+2C_LR_L}{2C_4C_LR_4R_L}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $\frac{C_LR_4R_L}{2C_4R_4+C_LR_4+2C_LR_L}$
Qz: None
Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(\infty,\ \infty,\ \infty,\ R_4 + \frac{1}{C_4 s},\ \infty,\ \frac{R_L}{C_L R_L s + 1}\right)$

$$H(s) = \frac{C_4 R_4 R_L s + R_L}{C_4 C_L R_4 R_L s^2 + s\left(C_4 R_4 + 2 C_4 R_L + C_L R_L\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_4}\sqrt{C_L}\sqrt{R_4}\sqrt{R_L}}{C_4R_4+2C_4R_L+C_LR_L}$
wo: $\frac{1}{\sqrt{C_4}\sqrt{C_L}\sqrt{R_4}\sqrt{R_L}}$
bandwidth: $\frac{C_4R_4+2C_4R_L+C_LR_L}{C_4C_LR_4R_L}$
K-LP: R_L
K-HP: 0
K-BP: $\frac{C_4R_4R_L}{C_4R_4+2C_4R_L+C_LR_L}$
Qz: None
Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (\infty,\ \infty,\ \infty,\ R_4,\ \infty,\ R_L)$

$$H(s) = \frac{R_4 R_L}{R_4 + 2R_L}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(\infty,\ \infty,\ \infty,\ R_4,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_4}{C_L R_4 s + 2}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(\infty,\ \infty,\ \infty,\ R_4,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{R_4 R_L}{C_L R_4 R_L s + R_4 + 2R_L}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(\infty,\ \infty,\ \infty,\ R_4,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_L R_4 R_L s + R_4}{s\left(C_L R_4 + 2C_L R_L\right) + 2}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(\infty,\ \infty,\ \infty,\ \frac{1}{C_4 s},\ \infty,\ R_L\right)$

$$H(s) = \frac{R_L}{2C_4 R_L s + 1}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(\infty,\ \infty,\ \infty,\ \frac{1}{C_4 s},\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{1}{s\left(2C_4 + C_L\right)}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(\infty,\ \infty,\ \infty,\ \frac{1}{C_4 s},\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{R_L}{s\left(2C_4 R_L + C_L R_L\right) + 1}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(\infty,\ \infty,\ \infty,\ \frac{1}{C_4 s},\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_L R_L s + 1}{2C_4 C_L R_L s^2 + s\left(2C_4 + C_L\right)}$$

9.9 X-INVALID-ORDER-9 $Z(s) = \left(\infty,\ \infty,\ \infty,\ \frac{R_4}{C_4 R_4 s+1},\ \infty,\ R_L\right)$

$$H(s) = \frac{R_4 R_L}{2C_4 R_4 R_L s + R_4 + 2R_L}$$

9.10 X-INVALID-ORDER-10 $Z(s) = \left(\infty,\ \infty,\ \infty,\ \frac{R_4}{C_4 R_4 s+1},\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_4}{s\left(2C_4R_4 + C_LR_4\right) + 2}$$

9.11 X-INVALID-ORDER-11 $Z(s) = \left(\infty,\ \infty,\ \infty,\ \frac{R_4}{C_4 R_4 s+1},\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{R_4R_L}{R_4 + 2R_L + s\left(2C_4R_4R_L + C_LR_4R_L\right)}$$

9.12 X-INVALID-ORDER-12 $Z(s) = \left(\infty,\ \infty,\ \infty,\ R_4 + \frac{1}{C_4 s},\ \infty,\ R_L\right)$

$$H(s) = \frac{C_4R_4R_Ls + R_L}{s\left(C_4R_4 + 2C_4R_L\right) + 1}$$

9.13 X-INVALID-ORDER-13 $Z(s) = \left(\infty,\ \infty,\ \infty,\ R_4 + \frac{1}{C_4 s},\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_4R_4s + 1}{C_4C_LR_4s^2 + s\left(2C_4 + C_L\right)}$$

9.14 X-INVALID-ORDER-14 $Z(s) = \left(\infty,\ \infty,\ \infty,\ R_4 + \frac{1}{C_4 s},\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_4C_LR_4R_Ls^2 + s\left(C_4R_4 + C_LR_L\right) + 1}{s^2\left(C_4C_LR_4 + 2C_4C_LR_L\right) + s\left(2C_4 + C_L\right)}$$

10 X-INVALID-WZ

11 X-PolynomialError